EXPERIMENT – 1

AIM :

To write a Python program to solve the 8-Puzzle Problem using a search algorithm.

ALGORITHM:

1. Start.
2. Initialize the initial state and goal state of the puzzle.
3. Find the blank tile (0).
4. Generate possible states by moving the blank tile up, down, left, or right.
5. Select the next state using a search algorithm.
6. Repeat until the goal state is reached.
7. Display the solution path.
8. Stop.

PROGRAM :

Experiment 1.py - D:/Abhi/Collage/AI/AI-Lab experiment-2026/Experiment 1.py (3.14.3)

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```
start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

def solve(start, goal):
    queue = deque([(start, [])])
    visited = {start}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        zero = state.index(0)
        row, col = divmod(zero, 3)

        moves = [(-1, 0), (1, 0),
                 (0, -1), (0, 1)]

        for dr, dc in moves:
            r, c = row + dr, col + dc

            if 0 <= r < 3 and 0 <= c < 3:
                pos = r * 3 + c
                new_state = list(state)
                new_state[zero], new_state[pos] = (
                    new_state[pos], new_state[zero]
                )
                new_state = tuple(new_state)

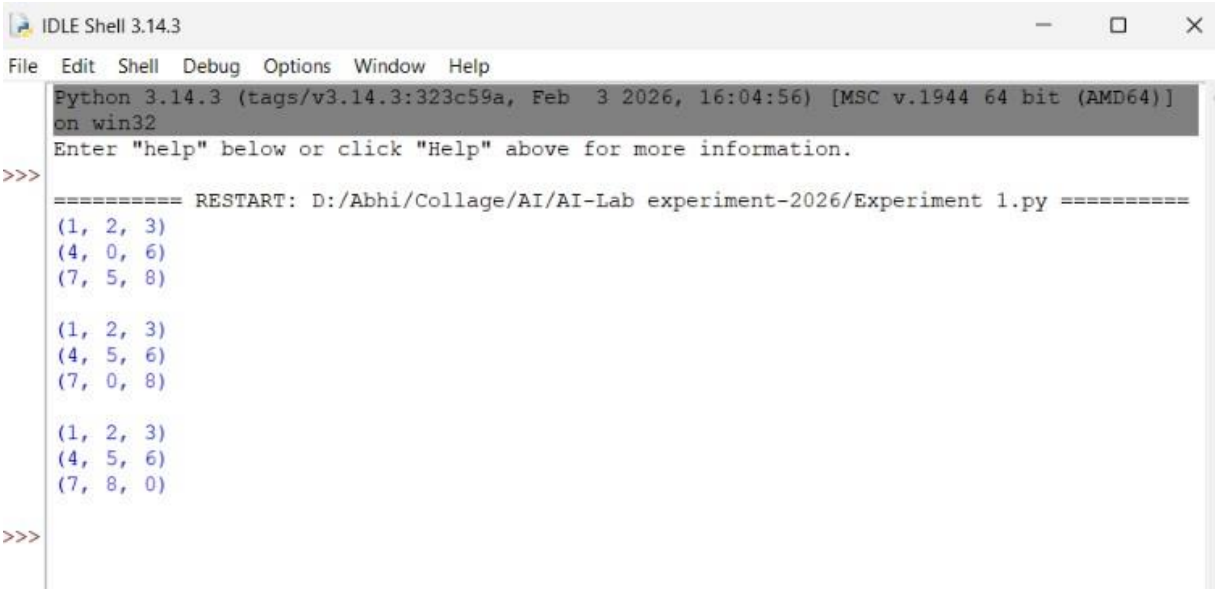
                if new_state not in visited:
                    visited.add(new_state)
                    queue.append(
                        (new_state, path + [state])
                    )

    return None

solution = solve(start, goal)

if solution:
    for state in solution:
        for i in range(0, 9, 3):
            print(state[i:i+3])
        print()
else:
    print("No solution found")
```

OUTPUT:



RESULT:

Thus, the python program to solve the 8-puzzle problem was executed successfully, and the goal state was reached.